

Digital vs. Analog Signals

Date: 2026-07-26

Cue / Keyw ords	Notes
Digital vs Analog	- Analog Systems/Signals: Continuous, smooth physical quantities (waves/voltages). Infinite values. Used in traditional tech (radio/TV). - Digital Systems/Signals: Discrete steps/numbers (binary 0s/1s). Finite set of values. Used in modern tech (computers).
Transmission Methods	- Analog Transmission: Continuous wave form. Pros: Simplicity, signal fidelity, real-time processing. - Digital Transmission: Discrete/binary form. Pros: High accuracy, efficiency, multiplexing, compression, security.
Signal Conversion	- ADC (Analog-to-Digital Converter): Converts real-world analog signals (sensors/audio) to binary for digital circuits/computers. - DAC (Digital-to-Analog Converter): Converts digital binary back into continuous analog voltage levels.
Modulation Techniques	Modulation adds information to a constant carrier signal for transmission. - Analog Modulation: Varies Amplitude (AM), Frequency (FM), or Phase (PM). - Digital Modulation (Shift Keying): Shifts Amplitude (ASK), Frequency (FSK), or Phase (PSK) to represent digital data.

Summary

Data communication relies on both analog (continuous) and digital (discrete) signals and transmission methods. Because the real world is analog and computers process digital data, ADCs and DACs are essential to bridge the gap. For transmission over distances, modulation is used to alter the amplitude, frequency, or phase of a carrier signal using either analog techniques or digital shift keying.